When does better AI increase token demand?

Adoption, delegated scope, and scarce human attention

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Abstract

Better AI can expand the amount of work assigned to AI while reducing the tokens used per work unit. We provide a mechanism-based decomposition of those two responses. When worthwhile work is finite, the expansion margin is adoption. When worthwhile work is abundant but human attention binds, it is work supervised per review hour. The active constraint identifies what must expand for token demand to rise; it does not determine whether expansion outweighs inference savings. Users choose delegated scope and inference intensity, and every attempt consumes tokens and review time, including failures. Two conditional diagnostics complement this framework. Under a pure efficiency change, the spending response to an equivalent price cut identifies the token-demand response. When attention binds, uniformly faster verification expands token demand and completed work in proportion to the capacity released. The numerical examples illustrate these mechanisms; they are not industry estimates or aggregate forecasts.

1. The economic question

Suppose a retailer is considering optional improvements to the descriptions and metadata in a fixed product catalog. A better model might produce each usable revision with fewer tokens. If most eligible listings already use AI, token purchases could fall. A rejected revision leaves the existing listing in place but still consumes inference and editorial attention, so failure is costly without requiring repair inside the model.

Now suppose a software team has a long backlog of worthwhile, optional improvements but little engineering time to check proposed changes. A better model might let each reviewer handle larger coherent assignments, expanding the amount of work attempted. Token purchases could rise even if each work unit needs fewer tokens. Whether they do depends on how review time grows with the amount delegated.

The distinction is not the occupation. It is what limits activity: the amount of worthwhile work, or the attention available to supervise it. Either team can move from one situation to the other.

The accounting is simple:

$$\text{Token demand} = \text{work assigned to AI} \times \text{tokens per work unit.}$$

The economics determines both factors. Higher capability can bring work into use, permit longer assignments, or save inference on existing work. Lower prices encourage extra inference. Greater efficiency supplies the same effective computation with fewer tokens. None of these responses is captured by assigning AI a fixed share of the value of the work it touches.

Efficiency rebound is not new. The energy literature relates efficiency changes to the demand for useful work, input demand, and price elasticities, while emphasizing the assumptions needed to infer one response from another (Sorrell and Dimitropoulos 2008). The organizational-economics literature asks how scarce expert knowledge and communication shape the span of production (Garicano 2000). This paper connects those ideas at the workflow level by letting users choose inference intensity and delegated scope together. It models review rather than an organizational hierarchy, and direct workflow demand rather than economy-wide rebound.

The contribution can be stated in one sentence: **the constraint identifies the expansion margin, while its size relative to inference savings determines whether token demand rises**. Under finite work, expansion means adoption; under scarce attention, it means supervisory leverage - work supervised per review hour. Price experiments and uniform-review improvements then provide conditional diagnostics. Capability has no universal quantity effect because it changes both expansion and inference intensity.

The argument starts with user choices and the two constraints. We then study capability, price and efficiency, and verification. Four figures show the mechanisms. Functional forms, proofs, parameter comparisons, and additional experiments are in the appendices.

2. A model of delegated work

2.1 Two choices and one review

A work unit is the amount of useful work a reference human would perform in an hour. This is a unit of task scope, not the AI's running time. A user chooses:

- **Scope** s**:** work delegated before the next review.
- **Inference intensity** x**:** tokens spent per work unit.

A chunk therefore uses sx tokens. Larger scope amortizes checkpoints over more work but makes successful completion harder. More inference improves execution at a cost. Both choices are needed: fixing scope removes the supervision response, while fixing inference intensity removes token savings.

Let $P(s, \eta x; m)$ be the probability that a chunk succeeds. Capability m improves success at a given policy; efficiency η provides more effective inference per token. Success falls with scope and rises with effective inference. Review takes $h(s; v)$ human hours, where a lower v makes review uniformly faster. For now, no particular formula for either function is needed.

Every chunk is attempted once and reviewed once. A successful chunk produces s units of completed work; a failed chunk produces none during the modeled period. Both consume the full token budget and review time. With output value b per completed work unit, token price c, and review-hour opportunity cost w, expected surplus per assigned work unit is

$$u(s,x) = bP(s, \eta x; m) - cx - w\frac{h(s; v)}{s}.$$

Failure is costly under either constraint. It uses paid inference and human attention without creating output. No extra failure penalty is required to make reliability matter.

This formulation describes work that can be checked after an attempt. Review identifies success, its duration depends on scope rather than the outcome or the raw token count, and the user cannot identify infeasible chunks for free before assignment. Partial completion, repair, and undetected damage are excluded. These choices keep the mechanism small; they also limit the settings to which it applies.

2.2 What is scarce?

Limited work. There are W potential work units, and review time can be obtained at cost w. Work opportunities differ in their outside options and switching hurdles. A work unit adopts AI if optimized surplus $u^* = \max_{s,x} u(s,x)$ exceeds its nonnegative hurdle ϕ. Let $A = F(u^*)$ be the work-weighted adopted share.

Hurdles do not depend on the policy. Each opportunity therefore prefers the same (s, x) before deciding whether to adopt. This assumption separates adoption from policy choice; it does not model users selecting progressively harder tasks. The main results require only a nondecreasing hurdle distribution. The numerical distribution is specified in Appendix A.

Limited attention. Useful work is abundant, but the user has only H review hours. A policy launches $s/h(s)$ work units per review hour. Define this quantity as $L(s)$, or work supervised per hour. Conditional on operating, the user maximizes

$$J(s,x) = L(s)[bP(s,\eta x;m) - cx] - w = L(s)u(s,x).$$

This is expected surplus per scarce review hour. If $\max J \leq 0$, the user can choose not to operate; all plotted attention cases have positive operating value. Abundant work means there are enough opportunities at the modeled value and negligible additional switching hurdles to fill capacity.

The two objectives select different policies because an hour spent on one assignment can crowd out other useful work when attention binds.

	Limited work	Limited attention
Policy maximizes	Surplus per work unit, u	Surplus per review hour, J
Work assigned, Q	WA	HL
Token demand, $D = Qx$	WAx	HLx
Expected completed work, $Y = QP$	WAP	HLP

Tokens and review hours are counted for all assigned work, including failures. Completed work is a separate outcome. Token spending, $S = cD$, is separate again: it is supplier revenue, not supplier profit or the economic value created.

The catalog example maps to the limited-work case. W is the set of eligible revisions during the period, s is revision work delegated before an editorial checkpoint, x is inference per unit of revision work, and A is the adopted share. The hurdle ϕ can represent the value of keeping existing content, integration effort, or another switching cost. An unusable revision is discarded, so it creates no incremental output while still using tokens and editor time. The potential revisions are otherwise technically and economically comparable in this model; real catalogs need not be. The example is optional content improvement, not work that must be completed by a deadline.

The software example maps to the attention-limited case when a queue of comparable optional improvements can keep reviewers busy. H is engineering review time, s is the amount of work proposed before the next checkpoint, x is inference per unit of engineering work, and $h(s)$ is the time needed to assess the change. The example is deliberately narrow: a rejected change can be abandoned, review reliably identifies success, and the queue is deep relative to review capacity during the period.

We study these two limiting cases to expose the mechanisms. When both constraints matter, scarce review time changes the policy itself. Appendix A gives their common allocation foundation; applying an attention cap after choosing a work-optimal policy is generally insufficient.

3. Capability: adoption or supervision?

Capability improves what a given inference budget can accomplish. The old policy remains available, so optimized surplus cannot fall. Token demand can fall because users change the policy. The relevant question is whether activity expands faster than tokens per work unit decline.

For any improvement, $D = Qx$: demand changes through activity and inference intensity. The resource constraint tells us which activity response to investigate; the accounting identity is not itself a prediction.

3.1 Finite work: adoption eventually has less room to grow

With fixed W, assigned work is $Q = WA$. Demand rises when adoption growth outweighs any reduction in inference intensity. Once adoption is near its ceiling, additional capability has little work left to bring into use. Token savings can then dominate.

This gives a simple saturation test that does not depend on the numerical adoption curve. From an initial policy (A_0, x_0) to a new policy (A_1, x_1),

$$\frac{D_1}{D_0} = \frac{A_1}{A_0}\frac{x_1}{x_0} \leq \frac{1}{A_0}\frac{x_1}{x_0}.$$

More generally, if reoptimized inference intensity falls to a fraction r of its initial value, demand must fall whenever initial adoption exceeds r. Thus, if more than half of the fixed workload already uses AI and chosen intensity falls by half, total token demand must fall even if every remaining work unit adopts. This is a conditional diagnostic, not a claim that doubling technical efficiency mechanically halves chosen intensity.

Figure 1 shows both margins, not just their product. The fixed catalog is the motivating workflow: adoption is the share of eligible revisions assigned to AI, while inference intensity is tokens per unit of revision work. In these examples, inference intensity falls throughout the displayed capability range while adoption rises. Their interaction produces a hump in demand.

The concentration comparison isolates the adoption mechanism: the same technical response produces a sharper demand increase when many opportunities switch together. Early saturation moves the peak earlier. Beyond its baseline, that case completes more work while using fewer tokens.

This is one reason a demand elasticity estimated during adoption takeoff may not persist. The adoption distribution is unchanged in the experiment; the user moves through it. The same workflow can respond differently after most of its work has switched.

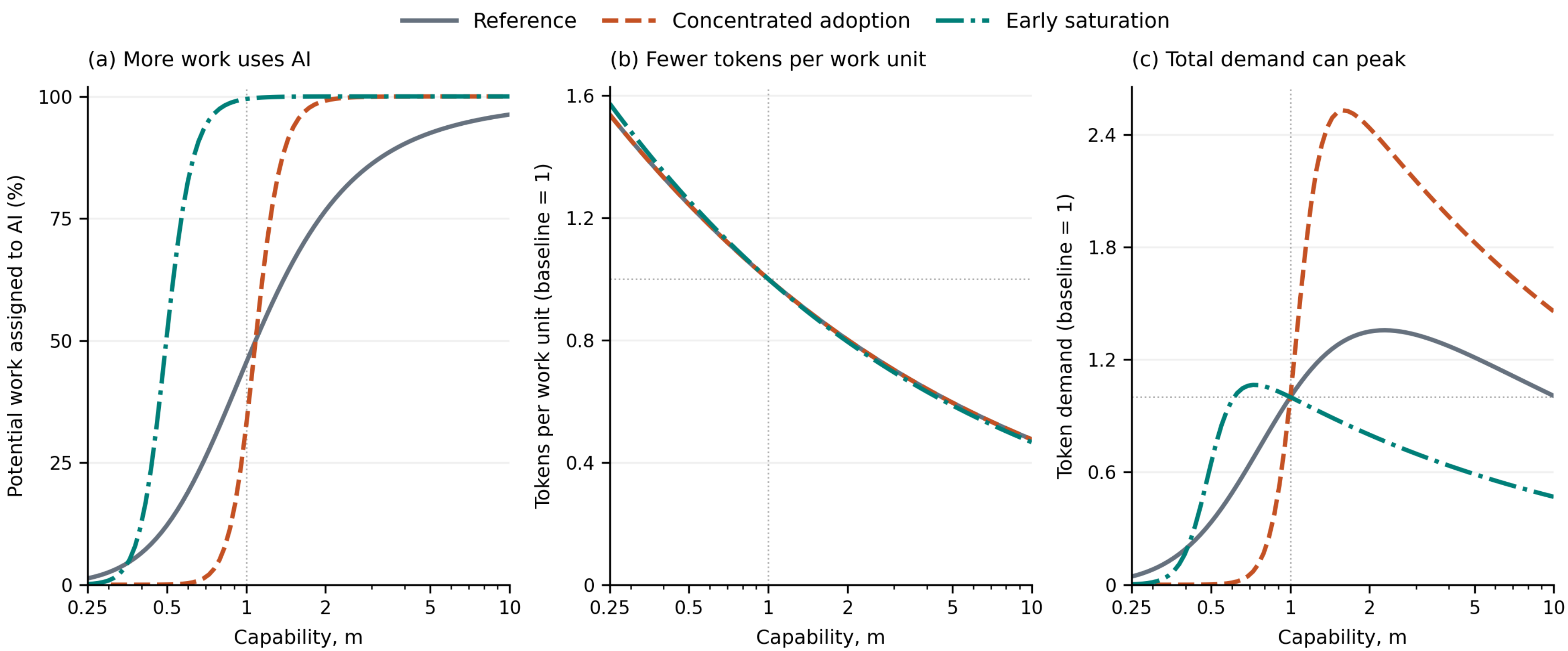
Neither an adoption plateau nor a capability score alone establishes a ceiling on token use. Users can still adjust inference, the workload can grow, and other workflows can enter. The plotted humps demonstrate a mechanism under fixed work, not an aggregate forecast.

3.2 Abundant work: review determines how far activity can expand

With fixed attention, assigned work is $Q = HL$. Larger chunks can increase $L = s/h(s)$ if review takes less than proportionately more time. If reviewing twice as much work takes almost twice as long, increasing scope releases little capacity.

In the software workflow, slowly growing review could describe a coherent change for which orientation and standard checks are substantially reusable as scope grows. Nearly proportional review could describe a change whose additional

Figure 1. Capability raises adoption while saving tokens per work unit; demand peaks when the balance changes.



The finite-work cases hold price, efficiency, and review technology fixed. Adoption is a percentage; the other panels use each case's own $m = 1$ *baseline. Reference and concentrated-adoption inference curves coincide because their technology and preferred policy are identical.*

components each require separate scrutiny. These are alternative review technologies for a stylized workflow, not claims that software review generally belongs to either case.

Figure 2 compares those two review technologies. It uses cases from the same parameter set, but adoption no longer constrains activity. The question is whether growth in work supervised per hour offsets inference savings.

When review grows slowly, the reviewer can supervise much more work and token demand rises strongly. With nearly proportional review, that expansion is modest. Inference savings eventually outweigh it, and token demand falls even though completed work and optimized value rise in this example.

The local object to measure is $\theta = d\log h / d\log s$, the percentage increase in review time associated with a one-percent increase in scope. An increase in scope raises work supervised per hour by $(1-\theta)$ times as much, in percentage terms. A long delegation horizon is economically useful only to the extent that it expands this ratio.

4. Price and efficiency: a conditional diagnostic

The preceding sections identify what must expand for demand to rise. Price variation provides a separate diagnostic for whether that expansion is large enough to produce direct efficiency rebound.

4.1 A price cut raises purchases, but need not raise spending

A lower token price makes additional inference worthwhile and raises the value of using AI. In the work-limited regime, chosen inference intensity and adoption both weakly increase. With attention limited, users weakly increase the tokens they buy per review hour. Thus a pure price cut cannot reduce optimized token quantity in either regime. This follows from revealed preference, without assuming a particular demand curve.

Spending can nevertheless fall. Halving price raises spending only if token purchases more than double. For a small price change, the threshold is a price elasticity of one: the percentage increase in quantity must exceed the percentage reduction in price.

Figure 3 asks where that expansion can come from. The reference and concentrated-adoption cases share the same technology. They differ only in how closely work opportunities' switching hurdles are clustered. The early-saturation case has easier tasks and already assigns almost all work to AI at the baseline. All cases are illustrative.

In the concentrated case, halving the baseline price moves adoption from roughly one-third to four-fifths. Token purchases rise about fourfold, so spending rises about twofold. Once most work has adopted, that source of expansion shrinks. Purchases can continue increasing while spending peaks and falls.

The lesson is conditional: a low price is especially effective at expanding the market when much work is near its adoption threshold. A large stock of potentially automatable work is not enough; its distance from that threshold matters.

4.2 Applying direct-rebound logic to inference efficiency

An efficiency gain is different from a price cut: it also reduces the raw tokens required for the same effective inference. But the two are linked exactly in this model.

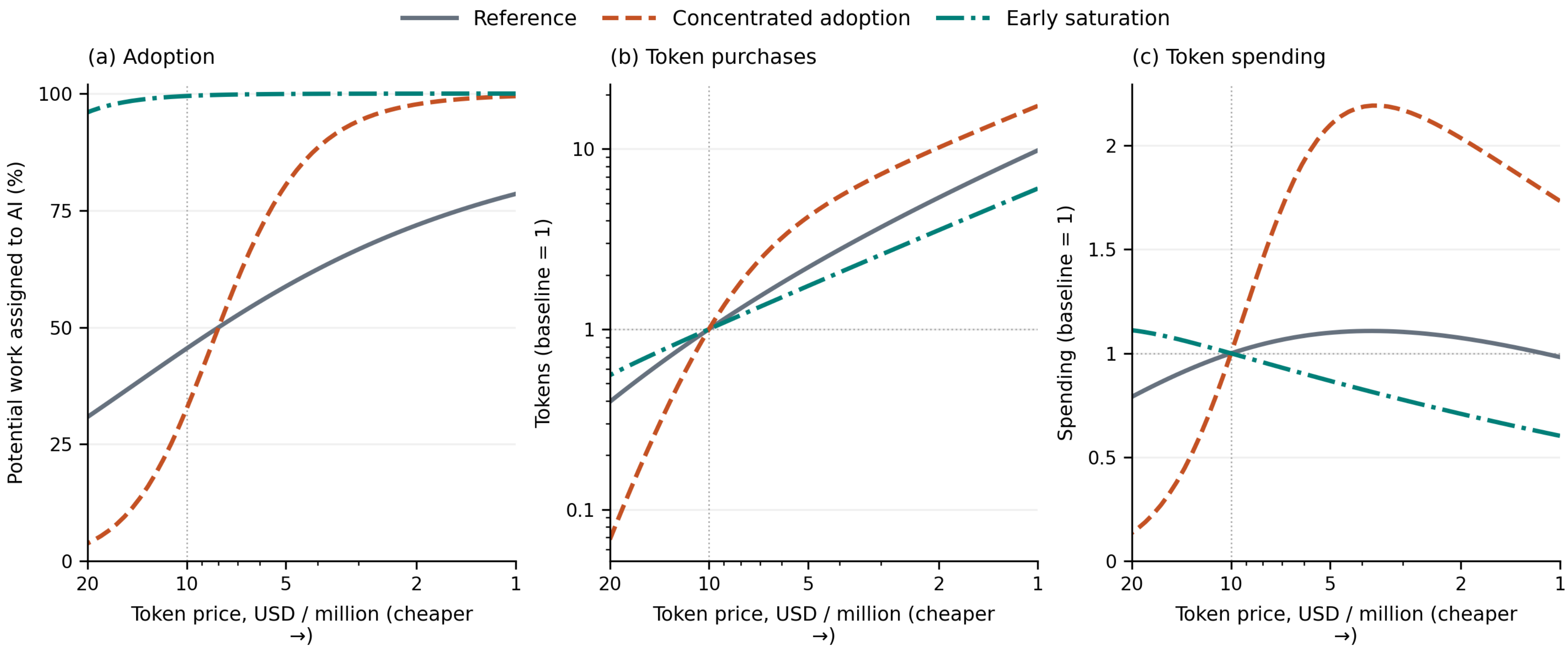
Write effective inference as $z = \eta x$. A policy using z costs $(c/\eta)z$ in tokens. Raising efficiency is therefore equivalent to lowering the price of effective inference, then dividing its use by the efficiency gain. Holding other inputs fixed,

$$D(c,\eta) = \frac{D(c/\eta, 1)}{\eta}.$$

Figure 2. Capability expands token demand when supervision scales; nearly proportional review can leave token savings dominant.

Attention is fixed and useful work is abundant. Every series uses its own $m = 1$ baseline. The first and third panels use logarithmic vertical scales. The cases have the same technical success function; their review technologies differ. Review levels scale demand, while the growth of review time with scope drives these indexed responses.

Figure 3. Cheaper tokens increase purchases; adoption determines whether spending keeps pace.



Prices fall from left to right. Adoption is a share of potential work; purchases and spending are indexed to each case's own value at 10 dollars per million tokens. The purchase panel uses a logarithmic vertical scale. Every point reoptimizes scope and inference intensity.

This is the direct-rebound relation specialized to inference. The local elasticity version is familiar from the energy literature; the following finite comparison lets a discrete price experiment reveal the corresponding efficiency response without estimating a local elasticity. For an improvement by any factor $k > 1$,

$$\boxed{\frac{D(c, k\eta)}{D(c, \eta)} = \frac{S(c/k, \eta)}{S(c, \eta)}.}$$

Doubling efficiency changes token purchases by the same proportion as halving price changes token spending. In the concentrated case above, a twofold efficiency gain therefore roughly doubles token use. In a case where halving price reduces spending, the same efficiency improvement reduces token use.

Equivalently, let ε be the positive price elasticity of token demand, evaluated at the corresponding effective price. Locally,

$$\text{elasticity of token demand to efficiency} = \varepsilon - 1.$$

Efficiency produces more raw token use only when effective demand expands more than proportionately. That is the rebound threshold. The result does not require a logistic adoption curve or the success function used in our examples.

It does require efficiency to enter through ηx, with no separate change in feasibility, review requirements, or available policy choices. A model release that changes all of these is not a pure efficiency experiment. Subject to that restriction, price variation can reveal the sign of rebound without separately estimating every technical parameter.

5. Verification expands the usable market

A faster model does not necessarily make its output faster to check. We therefore treat review technology as a separate margin. **Figure 4 asks what happens when every review takes less time**, holding capability, token efficiency, and price fixed.

For the software team, this means improving the checking process itself while keeping the pool of proposed work and model behavior fixed. For the catalog, it means reducing editorial time per attempted revision. The intervention is deliberately stronger than changing the review burden of only some tasks: it multiplies every policy's review time by the same factor.

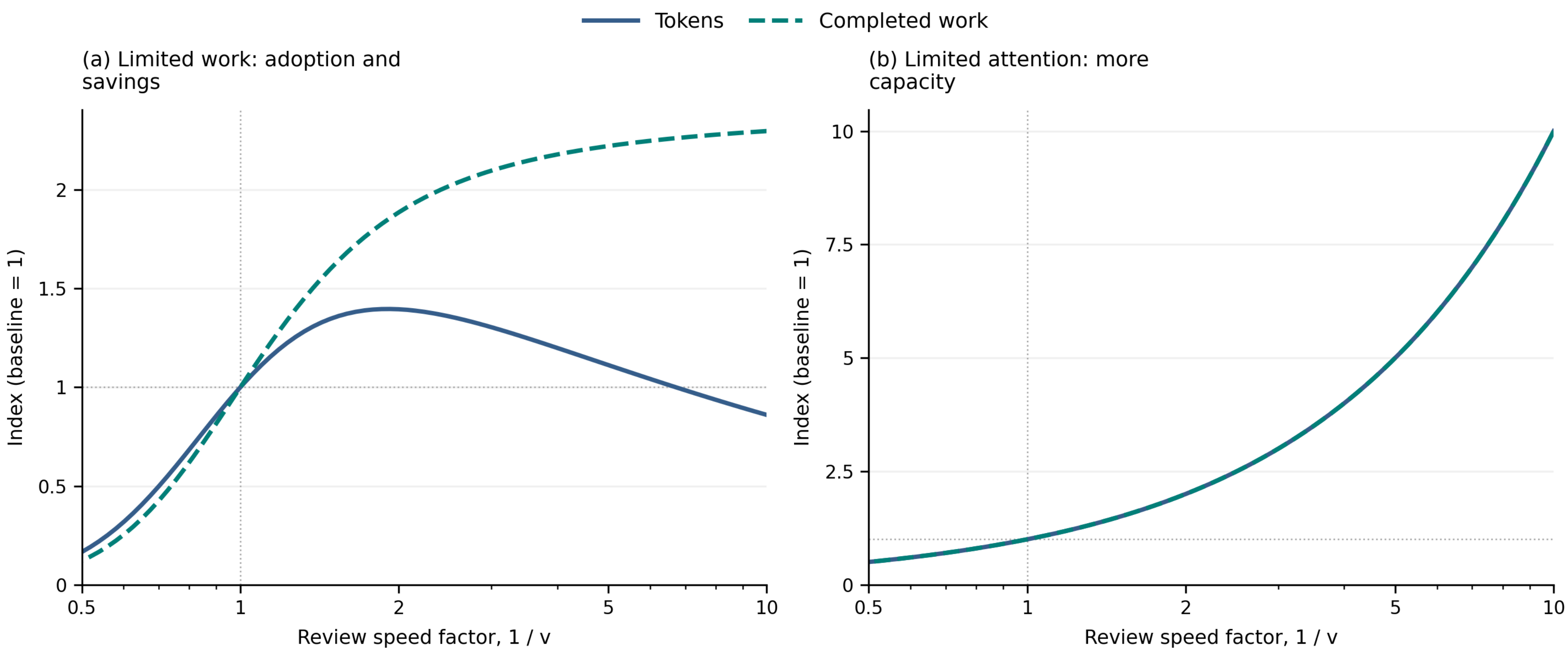
Under binding attention, write $h(s; v) = vh(s; 1)$. Lowering v multiplies the value of every policy per review hour by the same factor, before the fixed hourly charge. It leaves the preferred scope, inference intensity, and success probability unchanged. If useful work remains abundant,

$$D^H(v) = \frac{D^H(1)}{v}, \qquad Y^H(v) = \frac{Y^H(1)}{v}.$$

Halving review time doubles token demand and completed work. This result comes from releasing capacity; it does not require improved success.

With a finite workload, faster review instead changes policy and adoption. It does not mechanically create more catalog revisions once the eligible set is exhausted. In the reference example, halving review time nearly doubles completed work but raises token use by only about 40%. Larger speed gains eventually reduce token demand as adoption saturates and users save inference. Faster verification can therefore be complementary to aggregate token use in one regime and lead to net token savings in another.

Figure 4. Faster review can expand completed work more than token use; with scarce attention it expands both proportionally.



A review speed factor of two means half the review time at every scope. Tokens and completed work are indexed to the reference case's own baseline in each regime. In the attention panel the two curves coincide. Available work is fixed only in the left panel.

The proportional attention result stops applying when the extra capacity runs out of worthwhile work. A lower wage also has a different effect from faster review: it changes the monetary cost of operating, but does not create additional review hours. Conditional on activity in the attention-limited regime, w affects participation rather than the optimal policy. Changing how review grows with scope is another distinct intervention because it changes which policy is best instead of multiplying every policy's capacity equally. Appendix B examines that case and explains why lowering a review-growth exponent is not a uniform improvement for every task size.

6. Workflow regimes and what to measure

The model suggests classifying workflows by observable constraints rather than assigning an elasticity to an occupation or industry. The same organization can contain several regimes or move between them as adoption and review capacity change.

Hypothetical workflow	Diagnostic condition	Conditional demand implication
Optional revisions to a fixed catalog	How much eligible work remains near its switching threshold?	Adoption can drive early expansion, but loses force near saturation
Optional software improvements with reusable orientation and standard checks	Does review grow substantially less than proportionately with delegated scope?	Work supervised per review hour can expand enough to outweigh inference savings
Optional software improvements requiring separate scrutiny of each component	Does review grow almost proportionately with delegated scope?	Output can rise while token demand stagnates or falls

These are mechanisms to test, not empirical descriptions of retail or software as a whole. For the catalog, old content provides a natural outside option and the eligible set supplies a workload boundary. For software, the model applies only to a queue of optional, comparable changes that can be discarded after review. Mandatory repairs, task-dependent difficulty, partial reuse, and learning across attempts require additional structure. Review-tool compute and maintenance costs also lie outside $h(s)$ as currently specified.

Three measurements would make the framework useful in practice. First, **test which constraint is active**. Does more review capacity lead to more worthwhile work, or are existing review hours already sufficient? At the work-optimal policy, required attention is $WA\,h(s)/s$. If available hours are below that amount, the policy must be reconsidered. Faster-review experiments can help separate a supervision limit from limited adoption, provided the intervention does not change other task attributes. Within an attention-limited workflow, compare review time across assignment sizes to estimate whether larger scope actually releases capacity.

Second, **record scope, inference, and review time together**. A drop in tokens per completed task alone is hard to interpret: it can reflect smaller assignments, lower intensity, or higher success. Record assigned and successfully completed work separately, and include failed attempts in tokens and review hours.

Third, **measure the spending response to price**, holding model quality fixed and allowing users to adjust. That identifies the local rebound condition if an efficiency gain preserves the same effective-inference technology.

These measurements are more directly informative about token purchases than the dollar value of exposed work. They also separate economic progress from supplier revenue: more completed work can coexist with lower token use, and rising token use can coexist with lower spending.

The model is static. It holds the workload, attention endowment, output values, and the raw-token price fixed except for the input explicitly varied. New products, additional investment, organizational learning, and endogenous inference supply can move several margins together. Those responses must be added before interpreting a path as an aggregate forecast. Different models and token types also need a common inference unit; a physical token count is not automatically a comparable measure of compute.

7. Conclusion

The same improvement can expand the market for inference and reduce the inference needed for existing work. Standard direct-rebound logic identifies the balance through an elasticity. This paper gives that elasticity two operational sources in delegated AI work: adoption headroom when work is limited, and supervisory leverage when human attention is limited. It also gives two conditional diagnostics: the spending response to a price cut identifies the token response to a pure efficiency gain, and uniformly faster review scales active attention-limited demand with the capacity released. Neither result supplies a universal sign for capability.

A useful demand forecast should identify which work can enter, whether review binds, how review scales with scope, and how inference per work unit changes. The fixed catalog and optional software backlog make those questions concrete without assigning an elasticity to an entire industry. A fixed share of economic value cannot answer them, and a falling token bill is not sufficient evidence that AI has stopped becoming useful.

References

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- Sorrell, Steve, and John Dimitropoulos. 2008. “The Rebound Effect: Microeconomic Definitions, Limitations and Extensions.” *Ecological Economics* 65 (3): 636-649. Publisher page.

Appendix A. Specification and analytical results

A.1 Numerical technology and adoption

The main argument uses one success function. For the numerical examples we choose

$$P(s,x;m,\eta)=\exp\left[-\left(\frac{s}{\lambda m}\right)^{\nu}-\frac{s}{am[\eta(x/x_{\mathrm{ref}})]^{\alpha}}\right].$$

The first term is a capability limitation that more inference cannot remove at fixed scope. The second is an execution error that inference can reduce. Thus, as $x\to\infty$, success approaches $\exp[-(s/(\lambda m))^{\nu}]$, which can remain below one. The code retains the corresponding factors q and r for diagnostics; no additional user decision is associated with them.

The parameters λ and a set the two technical horizons. The exponent $\nu>0$ controls frontier shape, while $0<\alpha<1$ gives diminishing growth of the execution horizon in effective inference. This does not imply globally concave success probability or a universally concave optimization problem.

Here m proportionally extends both horizons. It is a specified path of technical improvement, not an observed intelligence scale or a claim that all model releases improve every task equally. Increasing λ alone instead isolates a feasibility improvement. These distinctions matter for capability comparisons; the general price and efficiency results do not depend on them.

Review is

$$h(s;v)=v(h_0+h_1 s^{\beta}).$$

The term h_0 is time spent opening, orienting to, and closing a checkpoint; $h_1 s^{\beta}$ is time spent examining its content. Scope s is expressed in units of one reference work hour, so h_1 is the variable review time at that scope. The numerical cases have $h_0>0$ and $0<\beta<1$. The local elasticity is

$$\theta(s)=\frac{sh_s}{h}=\frac{\beta h_1 s^{\beta}}{h_0+h_1 s^{\beta}}.$$

A constant review time would remove the content burden. Exactly proportional review with no overhead would remove the capacity gain from larger assignments. Keeping both parts distinguishes these cases without another choice variable.

For adoption, start with a logistic distribution of location μ and spread $\sigma>0$, conditioned on a positive hurdle. Its CDF is

$$F(u)=\begin{cases}0, & u\leq 0,\\ \dfrac{1-\exp(-u/\sigma)}{1+\exp[-(u-\mu)/\sigma]}, & u>0.\end{cases}$$

No work adopts at nonpositive surplus. The parameter μ is the underlying location, not exactly the median after conditioning. A smaller σ concentrates work near a similar switching threshold; it does not uniformly make adoption harder. Work values and technical characteristics are otherwise common within a case. Adding task-dependent values or review costs would couple selection and policy and would require a richer allocation model.

A.2 Interior choice conditions

Let subscripts denote partial derivatives. In either regime, an interior inference choice satisfies $bP_x = c$. The scope conditions differ:

$$\begin{aligned} &\text{Limited work:} && bP_s = w\frac{sh_s - h}{s^2}, \\ &\text{Limited attention:} && (1-\theta)(bP - cx) + sbP_s = 0. \end{aligned}$$

The first balances reliability against review cost per work unit. The second balances reliability against the surplus produced by greater work per review hour. Both charge tokens and attention regardless of success.

The comparative statics presume finite optimal policies exist. These are necessary conditions for interior optima, and the numerical solver does not assume that every stationary point is optimal. The plotted policies have positive operating value and do not reach their action bounds.

A.3 Price and efficiency results

In the work-limited objective, write $u_c(s,x) = B(s,x) - cx$, where $B = bP - wh/s$ does not depend on c. Compare optimizers at prices $c_1 < c_2$. Adding their two optimality inequalities gives

$$(c_2 - c_1)(x_1 - x_2) \geq 0.$$

Thus $x_1 \geq x_2$. Optimized surplus and the nondecreasing adoption share also rise weakly, so WAx does too. In the attention objective, price multiplies Lx; the identical argument shows that tokens per review hour weakly increase as price falls.

For efficiency, substitute $z = \eta x$ into either objective. Success and review are unchanged, and token spending becomes $(c/\eta)z$. Corresponding policies therefore have the same scope, surplus, adoption or supervised work, and completed work, while raw tokens differ by $1/\eta$. This proves the demand identity and its finite spending counterpart in Section 4.

The correspondence requires unrestricted positive choices or nonbinding bounds under the change of units. Hard raw-token caps, review costs tied to raw token count, or simultaneous changes in output value can break it. Fees outside cD also mean total invoices need not satisfy the spending identity: it concerns the variable token charge. At price-induced policy switches, the finite identity still applies to corresponding optima even when a local elasticity does not exist.

A.4 Capability, review, and the value of attention

Define optimized value per review hour after token spending but before charging for that review hour:

$$R^* = \max_{s,x} \frac{s}{h(s)}(bP - cx).$$

The net value of additional attention in the active regime is $R^* - w$. In the numerical technology, success depends on capability and scope through s/m, so $mP_m = -sP_s$. At a differentiable interior optimum, the envelope theorem and the attention scope condition give

$$\frac{d\log R^*}{d\log m} = \frac{b\,mP_m}{bP - cx} = 1 - \theta(s^*).$$

With $0 \le \beta \le 1$, this lies between $1-\beta$ and one. The formula concerns gross attention value; subtracting w changes its elasticity. It also requires the specified horizon-scaling capability path and fixed review technology.

The no-overhead benchmark follows by substituting $s = my$ when $h_0 = 0$ and $0 \le \beta < 1$. Gross value becomes

$$\frac{m^{1-\beta}}{vh_1} y^{1-\beta}[bP(y,x;1,\eta) - cx].$$

The maximizers of y and x do not depend on m. Hence $s^* \propto m$, x^* is constant, and $D^H \propto m^{1-\beta}$, provided choices remain interior. Fixed overhead removes this homogeneity and permits additional demand shapes; it is not needed for the general rebound result.

A.5 A common foundation when both constraints matter

Suppose work is divisible, technical characteristics are common, and review hours can be allocated across opportunities with different nonnegative hurdles. Charge a shadow price $\tau \ge 0$ for the shared attention constraint. The preferred policy and adoption surplus solve

$$\begin{aligned}(s_\tau, x_\tau) &\in \arg\max_{s,x} \left\{ bP - cx - (w+\tau)\frac{h(s)}{s} \right\}, \\ u_\tau &= \max_{s,x} \{ bP - cx - (w+\tau)h/s \}.\end{aligned}$$

Each work unit adopts if $\phi < u_\tau$. The scarcity price satisfies

$$WF(u_\tau)\frac{h(s_\tau)}{s_\tau} \le H, \qquad \tau\left[H - WF(u_\tau)\frac{h(s_\tau)}{s_\tau}\right] = 0.$$

These are the allocation conditions obtained by attaching the same attention price to each work opportunity's surplus net of its hurdle. When policies tie, a mixture can be needed to clear capacity; the attention use in the constraint is then the mixture's average. These conditions describe an efficient allocation under a common scarcity price, not a claim about arbitrary decentralized rationing.

Nonbinding attention gives $\tau = 0$ and the work-limited policy. If attention binds while W becomes very large and the hurdle distribution has support arbitrarily close to zero, marginal adopted hurdles approach zero. The limiting scarcity price is $\max J$, and the selected policy maximizes surplus per review hour. This gives the attention-limited case.

The computations report the two polar regimes. They do not numerically solve the full intermediate allocation. The evaluator's `realized_tokens` field is only capacity at a supplied policy; it is not that allocation solution or a participation decision.

Appendix B. Numerical examples and supplementary figures

B.1 Parameter choices and interpretation

One reference case, five high/low pairs, and three singleton configurations supply the comparisons. Each pair changes a related parameter group; singletons combine groups to illustrate additional possibilities. No case is an estimate of a named industry. Parameters appear below to make the examples reproducible, not to give their magnitudes empirical authority.

All cases use $w = 100$ dollars per review hour, $W = 1{,}000{,}000$ potential work units in work-limited comparisons, and $H = 100{,}000$ review hours in attention-limited comparisons. The scenario baseline is $m = \eta = v = 1$, $c = 10$ dollars per million tokens, and $x_{\text{ref}} = 100{,}000$ tokens per work unit. In equations, c is dollars per individual token.

The adoption location $\mu = 76$ is close to the reference work surplus of about 75.3, so the example starts near a transition. It is held fixed throughout. The reference policy assigns roughly half a work hour per chunk, succeeds about 91% of the time, uses about four minutes of review, and adopts about 45% of potential work. These are consequences of the chosen illustration, not measured averages.

Parameters run down the rows. **Bold cells differ from the reference.**

Technical conditions

Parameter	Reference	Frontier constraint low	Frontier constraint high	Execution difficulty low	Execution difficulty high	Review burden low	Review burden high
Capability horizon λ	12	**24**	**6**	12	12	12	12
Frontier shape ν	1.25	**1.5**	**1**	1.25	1.25	1.25	1.25
Execution ease a	4	4	4	**8**	**2**	4	4
Inference returns α	0.5	0.5	0.5	**0.65**	**0.35**	0.5	0.5
Fixed review h_0 (hours)	0.03	0.03	0.03	0.03	0.03	**0.015**	**0.06**
Variable review h_1 (hours)	0.05	0.05	0.05	0.05	0.05	**0.025**	**0.1**
Review growth β	0.5	0.5	0.5	0.5	0.5	**0.25**	**0.95**

Economic and adoption conditions

Parameter	Reference	Value low	Value high	Concentration low	Concentration high
Work value b (dollars)	100	**50**	**200**	100	100
Hurdle spread σ (dollars)	4	4	4	**16**	**1**
Hurdle location μ (dollars)	76	76	76	76	76

Singleton conditions

Parameter	Reference	Early saturation	Capability valley	Offsetting efficiency
Capability horizon λ	12	**36**	**36**	**36**
Execution ease a	4	**8**	4	**2**
Inference returns α	0.5	0.5	**0.25**	0.5
Fixed review h_0 (hours)	0.03	0.03	**0.001**	0.03
Review growth β	0.5	0.5	**0.9**	**0.8**
Hurdle spread σ (dollars)	4	**1**	4	4

The main figures shorten “Reference industry” to “Reference” and “Adoption concentration: high” to “Concentrated adoption.” The attention labels “Review grows slowly” and “Review nearly proportional” use the low- and high-verification-burden cases. Their overhead-to-scale ratios are identical, so the different indexed attention responses isolate review growth; the uniform differences in review level scale their unindexed quantities.

Technical high/low labels have the stated same-policy difficulty ordering over the plotted comparisons, which the notebook checks. Shape parameters need not preserve that ordering outside this range. Concentration CDFs cross near, rather than exactly at, μ after positive-hurdle conditioning.

For the concentrated price example, the exact displayed precisions are:

Price change	Adoption share	Token demand relative to baseline	Spending relative to baseline
10 to 5 dollars per million	32.6% to 80.4%	4.20	2.10

These values audit the prose example; none is a claimed estimate of a price response in a real industry.

B.2 Full parameter comparisons

The following figures retain all fourteen cases. Gray denotes the reference; colors group related comparisons, solid/dashed lines mark high/low variants, and dash-dot lines mark singletons. Level plots answer how much is purchased at the common illustrative endowment. Indexed plots answer how each case changes relative to its own baseline. They are different comparisons. These full sweeps span capability from 0.1 to 30, efficiency from 0.25 to 10, and price from 1 to 80 dollars per million, extending beyond the focused main-text views.

B.3 Additional capability and review shapes

The main argument does not require every response to be monotone or to have one peak. The next figure retains three selected attention cases, including the capability valley.

Setting fixed overhead to zero removes that valley in the homogeneous benchmark of Appendix A. The example therefore identifies which extra structure changes the response; it does not weaken the general price or efficiency results.

Figure B1. Work-limited demand in levels across the complete parameter set.

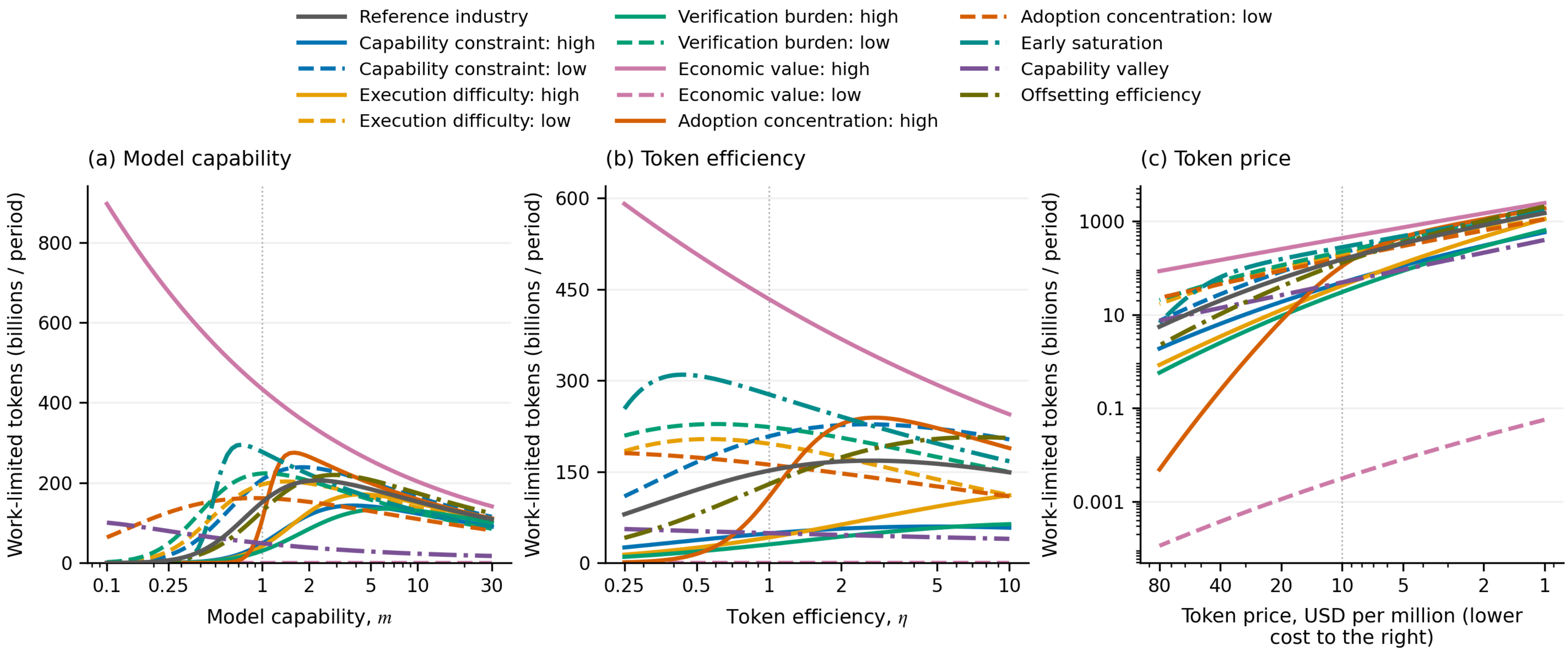
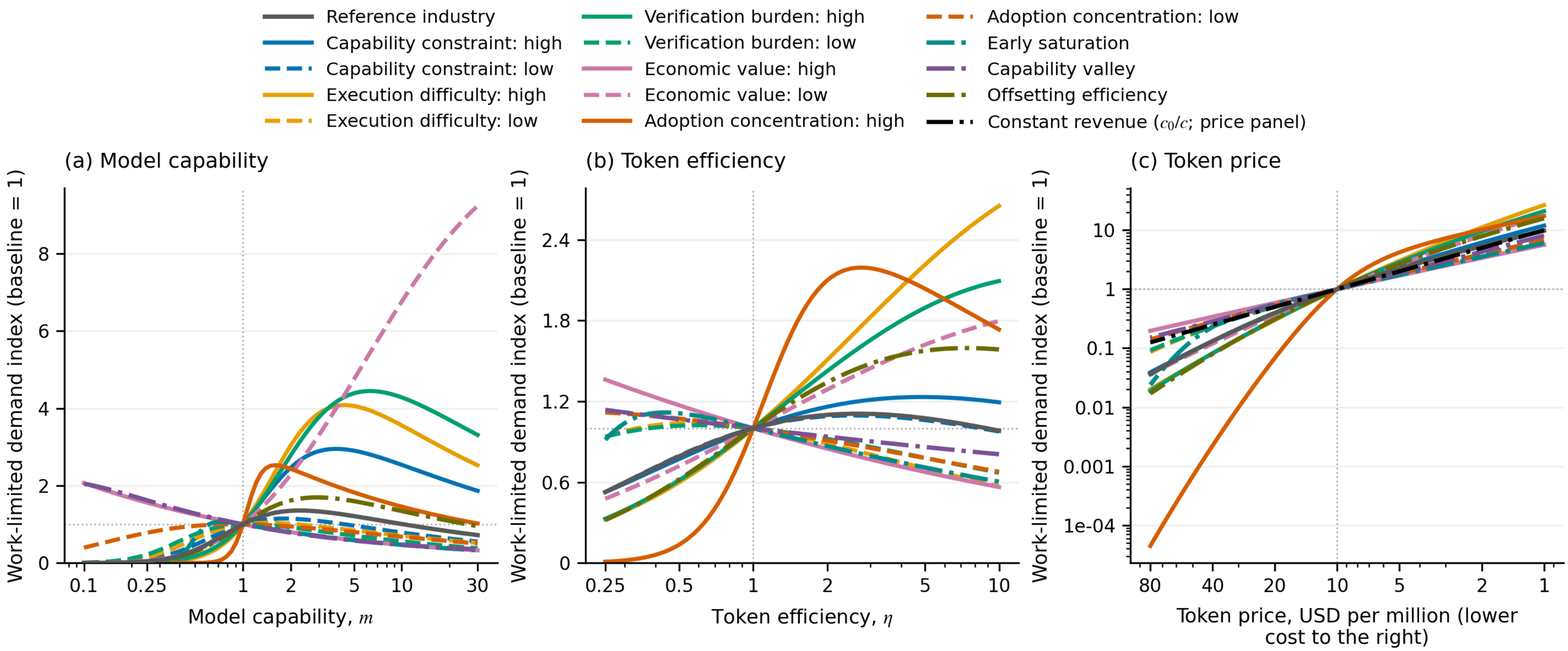


Figure B2. Work-limited responses indexed to each case's baseline.



Capability and efficiency panels use independent linear vertical scales; price uses a logarithmic scale. Price falls to the right. In the indexed price panel, the black c_0/c line is the quantity increase needed to preserve baseline spending. Being above it is a comparison with the baseline, not a claim that every further price cut increases spending.

Figure B3. Attention-limited demand in levels across the same parameter set.

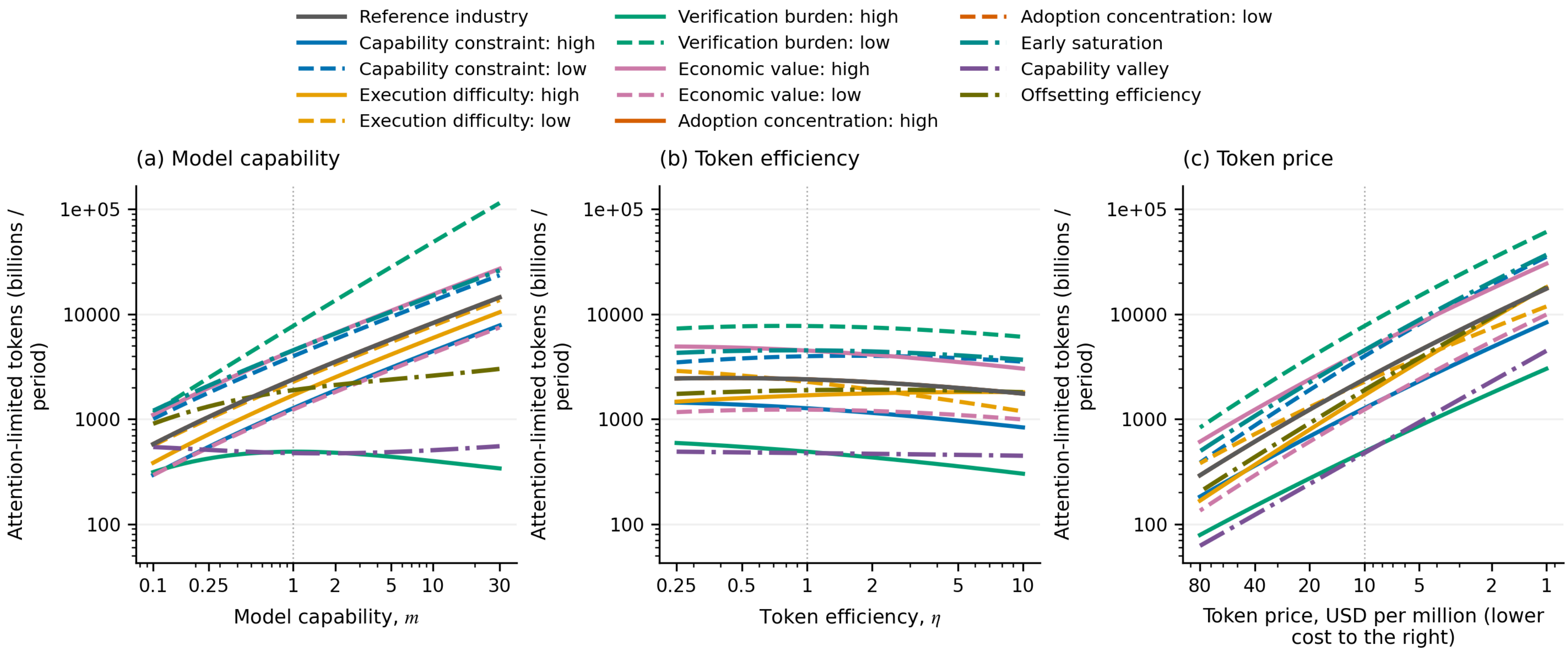
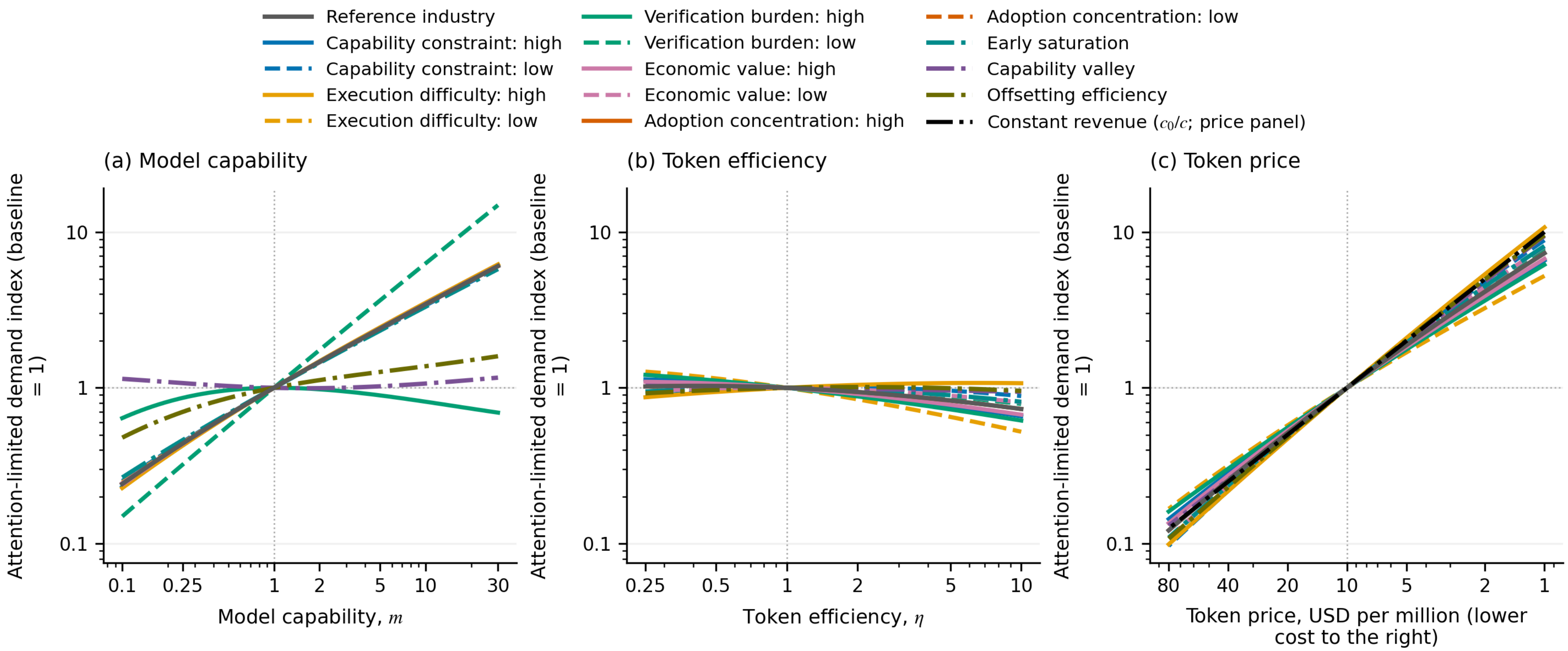
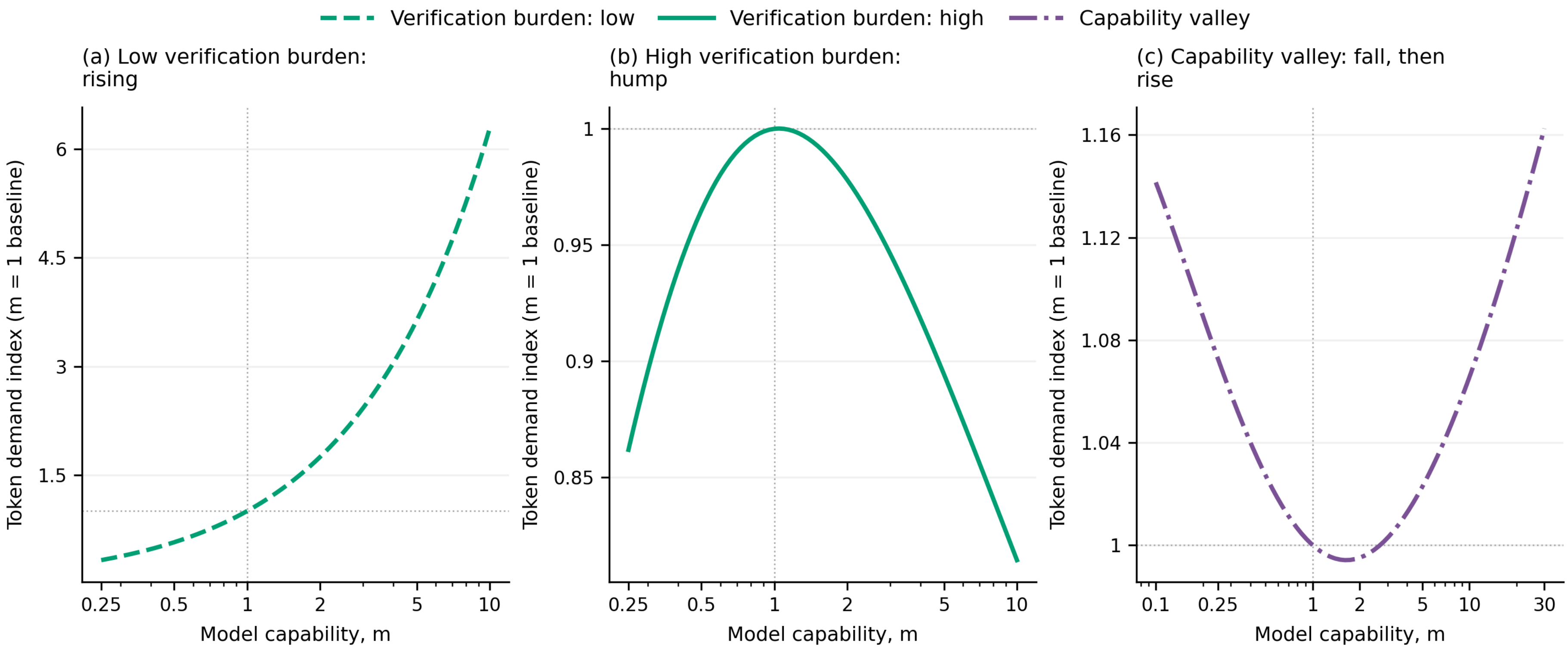


Figure B4. Attention-limited responses indexed to each case's baseline.



These figures use shared logarithmic vertical scales. The concentration cases coincide with the reference because switching hurdles do not constrain the abundant-work limit. A high demand level need not imply a high growth rate; review levels, endowments, and the starting point all affect levels.

Figure B5. Selected attention cases can rise, peak, or fall and recover as capability improves.



Each panel has its own vertical range. The valley is a selected combination of low fixed overhead, weak inference returns, and review growth close to proportional. Its reversal survives nearby parameter changes in the tests. This establishes a possibility, not its prevalence or an industry forecast.

The next experiment changes β, the growth of review time with scope, at capability levels $m = 1$ and $m = 5$.

The review functions cross at one work hour: lower β reduces review time for larger chunks but increases it for smaller ones. It is therefore not a uniform productivity improvement. In the reference work-limited case, preferred chunks are below one hour, and reducing β lowers adoption and completion. In the attention cases shown, it supports larger chunks and raises throughput. This contrast is a consequence of the intervention's crossing point, not evidence that uniformly faster verification reduces value.

B.4 Feasibility and large efficiency gains

A feasibility improvement raises λ alone; the next figure compares it with the same proportional increase in m.

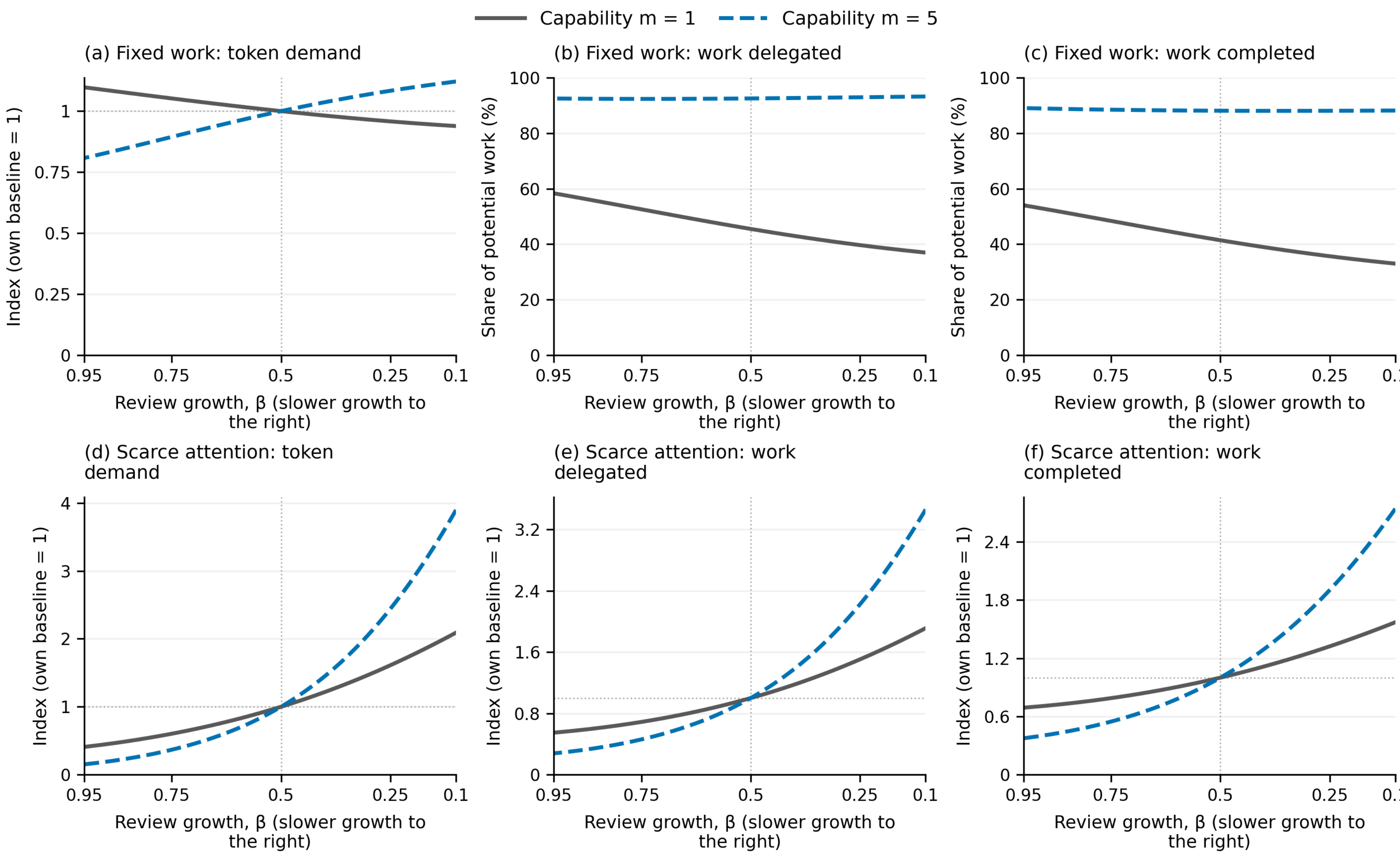
At a fivefold improvement, broader capability produces more completed work but a smaller increase in work-limited token demand than feasibility alone. Execution improvements save inference on work that becomes worthwhile. A better tool or harness could affect feasibility, execution, and review together; this experiment isolates only one of those channels.

Finally, we extend token efficiency to a hundredfold improvement at three inference-return settings.

Inference returns α	**Fixed-work demand at $\eta = 100$ (baseline = 1)**	**Scarce-attention demand at $\eta = 100$ (baseline = 1)**
0.25	1.23	0.86
0.50	0.59	0.43
0.75	0.29	0.22

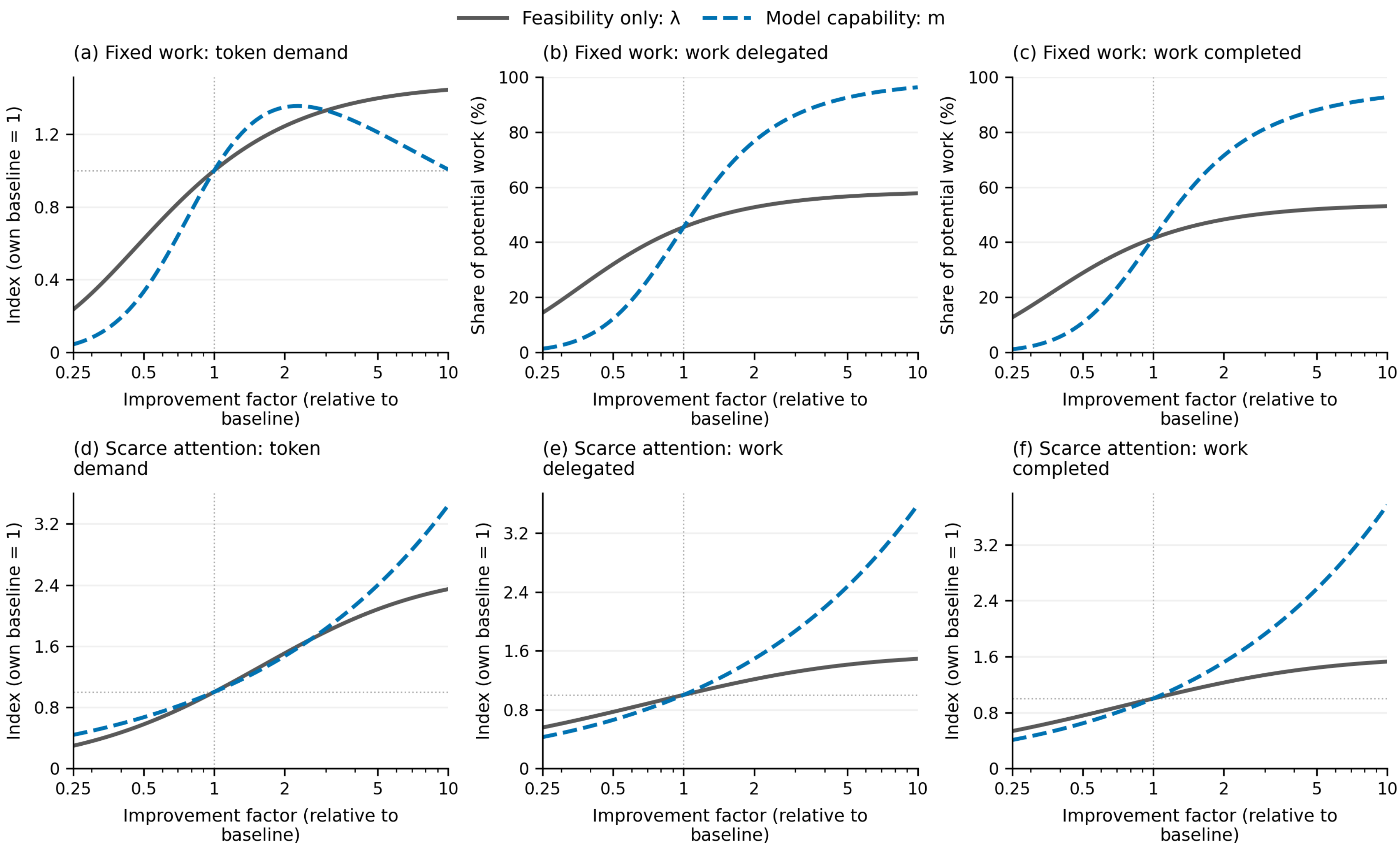
Completed work rises in every case shown. With weak inference returns, the work-limited example still uses more tokens at the endpoint because adoption expansion outweighs savings. The other work cases and all three attention cases use fewer tokens. A hundredfold efficiency improvement need not create a hundredfold reduction in token use, or any reduction at all. The ranking across α is conditional on the other illustrative parameters.

Figure B6. Changing review growth can affect small and large assignments differently.



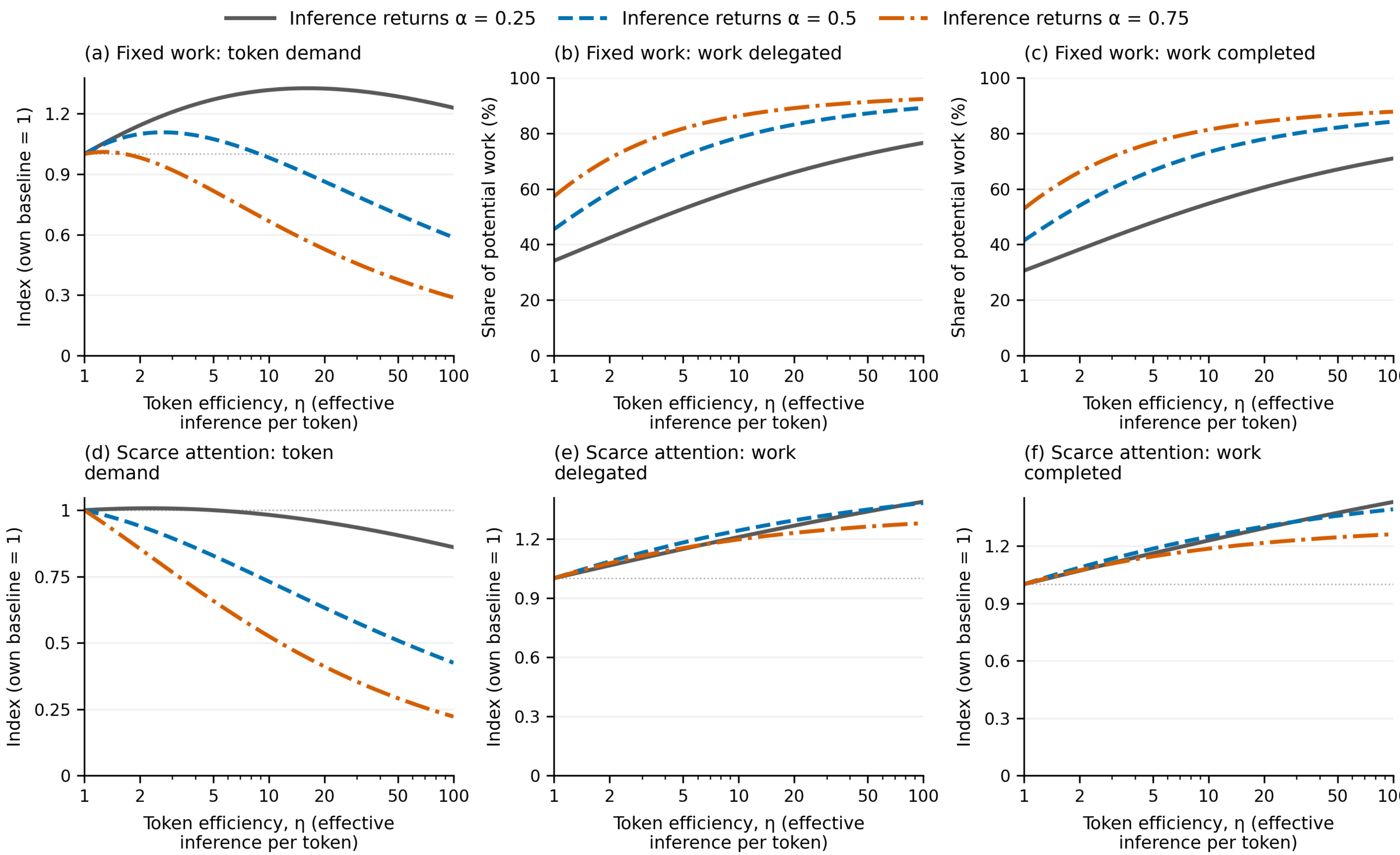
Lower β is to the right. The top row fixes work; the bottom row fixes attention. Columns show tokens, work assigned, and expected completed work. Work-limited assignment and completion are percentages of potential work; attention outcomes and token quantities are indexes.

Figure B7. Expanding feasibility and improving both technical horizons have different token consequences.



The two changes have the same effect on the first term of the success function at a fixed scope. Higher m also improves execution. Row and column meanings match Figure B6.

Figure B8. Large efficiency gains can increase completed work with very different token responses.



All curves use their own $\eta = 1$ baseline. Only efficiency varies along a curve. Inference-return exponents differ across curves; neither capability nor review technology changes. Row and column meanings match Figure B6.

Appendix C. Numerical checks and reproduction

The analysis notebook regenerates all 22 PNG figures, including ten additional views not placed in the manuscript. Its plotted coordinates also generate the interactive charts; the reading edition contains no separate demand model. The four main figures reuse audited outcomes from the full comparison and intervention data.

Both numerical optimizers search a 17×17 grid in $(\log s, \log x)$ and refine four starts. Bounds are $s \in [0.002, 800]$ and $x \in [200, 200{,}000{,}000]$. No plotted optimum reaches them. Independent checks use a 25×25 grid, eight starts, and tenfold wider bounds at endpoints, baselines, and sampled extrema.

The attention solution is also checked through an independent scalar characterization. Write $d = (s/(\lambda m))^{\nu}$, $t = s/[am(\eta x/x_{\text{ref}})^{\alpha}]$, and $\delta = 1 - \theta$. The interior conditions imply

$$cx = \alpha t b\, e^{-d-t}, \qquad t = \frac{\delta - \nu d}{1 + \alpha\delta}.$$

Substitution leaves one equation in s. The implementation rejects unsupported review technologies and solutions outside its configured bounds. Tests also check failure costs, nonnegative-hurdle adoption, both policy conditions, scarcity-price recovery of the attention policy, price-efficiency equivalence, and uniform-review scaling.

Most axes use 81 logarithmically spaced samples plus exact comparison anchors; review growth uses a linear grid. Lines join optimized samples without smoothing. Sampled peaks are not analytical turning points. Denser searches and alternative characterizations substantially reduce numerical concerns but are not a global proof for arbitrary parameter choices.

The comparison diagnostics and intervention diagnostics contain configurations, policies, unindexed outcomes, and independent audit errors. The self-review record documents the modeling and presentation reviews, including the comparison with the requested reference papers.

Install dependencies, run checks, and regenerate the reading edition:

```
uv sync --extra notebook --extra dev --extra paper
uv run pytest
node --test tests/paper_controls.test.cjs
uv run token-demand-paper refresh
```

For a prose-only rebuild with valid cached plots, use `uv run token-demand-paper build`. Output is `build/paper/index.html`. To open a local preview that refreshes as sources change:

```
uv run token-demand-paper serve --port 8001
```

Visit the local reading edition. Figures support line toggles, hover values, zooming, and expanded panels; static PNGs are used for printing and when scripts are unavailable. Full comparison plots remain available for readers who want to inspect parameter differences.

To share a static edition, copy the entire `build/paper/` directory. Plotly and chart data are packaged locally; equation typesetting uses a pinned MathJax CDN and needs internet access. Generated HTML is excluded from version control. The manuscript, figure sources, figures, and cached data are tracked. The preview serves generated files locally and does not publish the paper.